

Bloxd.io Encyclopedia

The Complete Reference — Every stat, every API, every example

98	232+	22	64	21	10
GAME MODES	SURVIVAL ITEMS	FIREARMS	CALLBACKS	BIOMES	MOB TYPES

SECTION I All Game Modes

1. Original Four (2021)

MODE	TYPE	DESCRIPTION
Bloxd Hop	Parkour	The first-ever mode. Speedrunning and parkour. Originally the entire game.
Evil Tower	Climbing	Escape a tower by climbing upward through obstacles.
Doodle Cube	Drawing	Creative drawing and building on a shared canvas.
Cube Warfare	FPS	Gun combat mode with 22 firearms. Added shortly after the OG Trio.

2. Open World Modes

MODE	DESCRIPTION
Peaceful	Open world with no hostile mobs. First open-world mode.
Creative	Unlimited blocks, free building, flying enabled.
Survival	Mine, craft, build, fight mobs. Hosts the largest item set (232+ items).
Worlds	Customizable lobbies with Code Blocks, multiple map types (Sky Grid, One Block Wide).

3. Current Active Modes

MODE	TYPE	VOTE RESULT
Bedwars 4v4v4v4	PvP	Squads beat Solos
Sky Wars	PvP	Beat Prison
Greenville	Roleplay	Beat OneBlock
One Block	Survival	Beat Murder Mystery
Hide and Seek	Party	Beat Duels
Murder Mystery	Party	Beat The Pit
Plots	Creative	Beat Prison
Pirates	PvP	Beat Survival Royale
Infection	PvP	Beat Walls
Rocket Spleef	Party	Beat Rush Racing
Rocket Obby	Parkour	Kept (no vote winner)
Lucky Towers	PvP	Kept (no vote winner)
Last Stand	Survival	Beat Goldrush
Natural Disaster	Survival	Beat Battle Box
99 Nights	Survival	Beat Sky Grid
Lucky Block Battlegrounds	PvP	Beat Pet Sim
Fishing	Casual	Beat Capture the Flag
Tsunami Steal Brainrots	Party	Beat Work at a Restaurant
The Bloxd SMP	Survival	Beat Simon Says
Dress Up	Cosmetic	Beat DimensionZ
Sky Islands	Survival	Beat Goal Rush
Core Defenders	Defense	Beat Heroes vs Brainrots
Bingo	Party	Stayed after Walls vote
Conqueror	PvP	Active
Custom Games	Various	Active

MODE	TYPE	VOTE RESULT
Clutch	Bridging	Active
Break Lucky Blocks for Aura	Casual	Active
Build Fight	PvP	Active
Bed Fight	PvP	Active
Bridge Fight	PvP	Active
1v1 Fights	PvP	Active
Battle Arena	PvP	Active

4. Survival Modes

- **Survival with Guns** — no vote needed (only contestant).
- **Day & Night Survival vs Build a Factory** — currently voting (TBA).

5. Legacy & Removed

MODE	LOST TO	STATUS
Duels	Hide and Seek	Removed
The Pit	Murder Mystery	Removed
Prison	Sky Wars, then Plots	Removed
Survival Royale	Pirates	Removed
Walls	Infection	Removed
Factions	Beat Anarchy	Deleted later
Anarchy	Factions	Removed
Rush Racing	Rocket Spleef	Removed
Defusal	Tie (both removed)	Removed
One Block Wide	Tie (both removed)	Worlds map type
Goldrush	Last Stand	Removed
Battle Box	Natural Disaster	Removed
Sky Grid	99 Nights	Worlds map type
Pet Sim	Lucky Block Battlegrounds	Removed
Capture the Flag	Fishing	Removed
Minigames	Bridge Battle	Legacy
Bridge Battle	Won vote	Legacy (Apr 2026)
Work at a Restaurant	Tsunami Steal Brainrots	Legacy
Simon Says	The Bloxd SMP	Legacy
DimensionZ	Dress Up	Legacy
Goal Rush	Sky Islands	Legacy
Heroes vs Brainrots	Core Defenders	Legacy

1. Pickaxes

Mine stone, ores, and most blocks. Enchantments: Break Speed, Momentum, Mining Yield, Mining Aura.

NAME	DMG	MINE SPEED	CRAFTING	MINABLE ORES	NOTES
Wood Pickaxe	18	1x	4 Wood Planks	Coal	Mines grass, dirt, stone, coal
Stone Pickaxe	22	1.3x	1 Plank + 3 Stone	Coal, Iron	+ iron ore
Iron Pickaxe	27	1.6x	1 Plank + 3 Iron Bar	Coal, Iron, Gold	+ gold ore, mines any ore
Gold Pickaxe	30	2x	1 Plank + 3 Gold Bar	All + Diamond	+ diamond ore
Diamond Pickaxe	32	2.5x	1 Plank + 3 Diamond	All + Moonstone	+ moonstone ore, obsidian
Moonstone Pickaxe	—	2.5x	1 Plank + 3 Moonstone	All	Drops exact blocks from ores (no smelting needed)
Golem Pickaxe	—	2.5x	3 Golem Eye + 3 Moonstone + 1 Messy Stone	All	Chance to drop bonus fragments

2. Spades

Dig dirt, sand, snow. Enchantments: Break Speed, Momentum, Digging Aura.

NAME	DMG	DIG SPEED	CRAFTING
Wood Spade	15	1x	2 Wood Planks
Stone Spade	19	1.3x	1 Plank + 1 Stone
Iron Spade	24	1.6x	1 Plank + 1 Iron Bar
Gold Spade	26	2x	1 Plank + 1 Gold Bar
Diamond Spade	28	2.5x	1 Plank + 1 Diamond

3. Hoes

Mine plants and leaves, till soil. Enchantments: Break Speed, Momentum, Farming Yield, Farming Aura.

NAME	DMG	CRAFTING
Wood Hoe	15	3 Wood Planks
Stone Hoe	19	1 Plank + 2 Stone
Iron Hoe	24	1 Plank + 2 Iron Bar
Gold Hoe	26	1 Plank + 2 Gold Bar
Diamond Hoe	28	1 Plank + 2 Diamond
Netherite Hoe	—	UNUSED (game files only)

4. Axes

Chop wood. Second-best melee weapons after swords. Enchantments: Break Speed, Momentum, Lumber Aura.

NAME	DMG	CHOP SPEED	CRAFTING	SPECIAL
Wood Axe	20	1x	4 Wood Planks	—
Stone Axe	24	1.3x	1 Plank + 3 Stone	—
Iron Axe	31	1.6x	1 Plank + 3 Iron Bar	—
Gold Axe	33	2x	1 Plank + 3 Gold Bar	—
Diamond Axe	36	2.5x	1 Plank + 3 Diamond	—
Artisan Axe	31	1.6x	Code Blocks only	Drops exact block
Moonstone Axe (Old)	36	—	Unobtainable	Source code only
Moonstone Axe (New)	36	2.5x	1 Plank + 3 Moonstone	Breaks 6 blocks above

5. Other Tools

NAME	DMG	CRAFTING	USE
Hands	4	Default	Break most blocks slowly
Stick	—	1 Plank	Increased knockback
Shears	27	1 Plank + 2 Iron Bar	Cut wool, leaves, flowers, grass
Compass	4	1 Iron + 1 Gold Bar	Points to coords / nearest enemy
Pebble	4	1 Messy Stone	Throwable projectile

SECTION III Weapons

1. Swords

Primary melee weapons. Enchantable with Damage I–III, Critical Damage, Vertical Knockback.

SWORD	BASE	CRIT	DMG I	DMG II	DMG III	CRAFTING
Wood	22	30	29	32	35	3 Planks
Stone	27	36	35	38	41	1 Plank + 2 Stone
Iron	34	46	43	47	50	1 Plank + 2 Iron
Gold	37	50	46	50	53	1 Plank + 2 Gold
Diamond	40	54	49	54	57	1 Plank + 2 Diamond
Knight	43	58	52	57	60	1 Diamond Sword + 3 Knight Hearts

2. Bows & Crossbows

Ranged weapons. Bows charge in 0.9s; crossbows in 1.15s but hold their charge indefinitely.

TIER	BOW DMG	BOW CHARGE	CROSSBOW DMG	CROSSBOW CHARGE
Wood	37	0.9s	37	1.15s
Stone	45	0.9s	45	1.15s
Iron	57	0.9s	57	1.15s
Gold	62	0.9s	62	1.15s
Diamond	68	0.9s	68	1.15s

3. Spears, Daggers, Clubs, Maces, Whips

Specialty weapons. Cannot be crafted normally—obtained via Code Blocks or special modes. Each has a Moonstone variant.

TYPE	WOOD	STONE	IRON	GOLD	DIAMOND	MOONSTONE	SPECIAL MECHANIC
Spear	18	22	28	30	32	32	Dash forward on attack
Dagger	10	12.5	15.5	17	18	18	2x backstab damage
Club	22	27	34	37	40	40	Extra knockback
Mace	11	14	17	19	20	20	Damage scales with player speed
Whip	8	11	14	16	17	17	Ranged melee (extended reach)

Boomerang also available in all tiers (throwable ranged weapon, returns to player).

4. Projectiles & Explosives

ITEM	DAMAGE	SOURCE	BEHAVIOR
Arrows	Varies (bow)	Craft: Stick + Stone	Ammo for bows/crossbows. Can carry potion effects
Pebble	4	1 Messy Stone	Throwable, single use
Blinding Pebble	4	Special	Blinds targets for 3s on hit
Firecracker Pebbles	12	Special	Small explosive radius on impact
Fireball	20	Moonstone craft	Arcing fire projectile, sets area alight
Bouncy Bomb	30	Special	Bounces 3 times before detonation
Frag Grenade	40	Special	3s fuse, large damage radius
Smoke Grenade	0	Special	Smoke cover for 8s, blocks vision
Mining Grenade	10	Special	Destroys blocks in 3x3x3 radius
Explosive (TNT)	60	Craft: Gunpowder + Sand	Large blast radius, destroys terrain
Mango	8	Fruity Mango Leaves	Throwable fruit with high knockback
Coconut	6	Fruity Palm Leaves	Gives speed boost to thrower when uncracked

SECTION IV Firearms (22 Guns)

Used in Cube Warfare and Survival with Guns. Not craftable in standard Survival.

GUN	TYPE	FIRE MODE	BODY DMG	HEAD DMG	MAGAZINE	FIRE RATE
AK-47	Assault Rifle	Auto	11	20	30	Fast
M16	Assault Rifle	Auto	10	18	30	Fast
SKAR	Assault Rifle	Auto	12	22	25	Medium
TAR-21	Assault Rifle	Auto	10	19	30	Fast
AWP	Sniper Rifle	Bolt-action	65	100	5	Slow
VSR	Sniper Rifle	Bolt-action	55	90	10	Slow
DMR	Marksman	Semi-auto	28	50	15	Medium
FMR	Rifle	Semi-auto	22	40	20	Medium
MP40	SMG	Auto	8	15	32	Very Fast
Mini Izu	SMG	Auto	7	13	35	Very Fast
M1911	Pistol	Semi-auto	15	28	7	Medium
Block-19	Pistol	Semi-auto	12	22	15	Fast
Deagle	Pistol	Semi-auto	32	55	7	Slow
One Shot Pistol	Pistol	Single	100	100	1	Very Slow
Double Barrel	Shotgun	2-shot	40	60	2	Slow
Striker-12	Shotgun	Semi-auto	25	40	12	Medium
GPMG	Machine Gun	Auto	9	16	100	Fast
Minigun	Heavy	Full auto	6	11	200	Very Fast
RPG	Launcher	Single	80	80	1	Very Slow
Super RPG	Launcher	Single	100	100	1	Very Slow
Grenade Launcher	Launcher	Arc	50	50	6	Medium

SECTION V Armor

1. Per-Piece Damage Reduction

Each set has 5 pieces: Helmet, Chestplate, Gauntlets, Leggings, Boots.

SET	HELMET	CHEST	GAUNTLETS	LEGS	BOOTS	TOTAL DR
Wood	5.97%	11.98%	4.8%	9.18%	4.1%	39.92%
Iron	7.98%	17.16%	6.63%	13.06%	5.97%	54.99%
Gold	9.43%	19.96%	9.18%	14.88%	7.98%	66.04%
Diamond	11.98%	22.49%	9.43%	17.68%	9.18%	75.47%
Fur	—	18%	—	—	0%	18%

Effective HP with armor: At 100 HP, Diamond armor gives an effective 408 HP against melee. Gold gives 294 HP. Wood gives 166 HP. Calculated as $HP / (1 - DR)$.

2. Colored Wood Armor

16 dye colors × 5 armor pieces = 80 total variants. Colors: White, Orange, Magenta, Light Blue, Yellow, Lime, Pink, Gray, Light Gray, Cyan, Purple, Blue, Brown, Green, Red, Black.

SECTION VI Food

Eat food to restore HP or gain temporary effects. Most foods stack up to 999. Eating speed matters—faster foods are better in combat.

FOOD	TIME	HP/EFFECT	HP/SEC	SOURCE	STACKABLE
Apple	0.75s	+12 HP	16.0	Fruity Maple Leaves	999
Bread	0.95s	+24 HP	25.3	Craft: 3 Wheat	999
Bowl of Rice	0.45s	+26 HP	57.8	Craft: 6 Rice + Bowl	999
Mushroom Soup	0.45s	+15 HP (+55 if HP<50)	33.3 / 122.2	Craft: Mushrooms + Bowl	999
Steak	1.25s	+28 HP	22.4	Smelt Raw Beef	999
Cooked Porkchop	1s	+25 HP	25.0	Smelt Raw Porkchop	999
Cooked Mutton	0.75s	+20 HP	26.7	Smelt Raw Mutton	999
Cooked Venison	1s	+16 HP	16.0	Smelt Raw Venison	999
Baked Potato	1.15s	+22 HP	19.1	Smelt Raw Potato	999
Cornbread	0.95s	+24 HP + 20 shield	25.3+	Unobtainable	999
Cherry	0.75s	Regen (5s)	~8/s	Fruity Cherry Leaves	999
Cracked Coconut	0.75s	Speed (8s)	—	Fruity Palm Leaves	999
Watermelon Slice	0.55s	Jump Boost (12s)	—	Forest / Seeds	999
Melon Slice	0.75s	Invisible (12s)	—	Forest / Seeds	999
Gold Watermelon	2s	Jump Boost (18s)	—	2 Slices + Gold Bar	999
Gold Melon	2s	Invisible (18s)	—	2 Slices + Gold Bar	999
Pear	1s	Dmg Reduction (13s)	—	Fruity Pear Leaves	999
Plum	1s	Damage I (8s)	—	Fruity Plum Leaves	999
Banana	1s	Double Jump	—	Gorillas / Seeds	999
Corn	1s	20 HP shield	—	Seeds / Meadow	999
Chili Pepper	0.75s	Heat Resist (30s)	—	Chili Plants	999
Carrot	0.8s	X-Ray (30s)	—	Carrot Plants	999
Beetroot	1s	Mining Yield (8s)	—	Beetroot Plants	999

FOOD	TIME	HP/EFFECT	HP/SEC	SOURCE	STACKABLE
Mango	1s	Knockback	—	Fruity Mango Leaves	999

Best combat food: Bowl of Rice (57.8 HP/sec) is the fastest healing food. Mushroom Soup is the best when low HP (122.2 HP/sec below 50 HP). For sustained fighting, stack Rice or Porkchops.

Raw meats (Porkchop, Beef, Mutton, Venison, Potato) inflict Poison for 4–5 seconds when eaten uncooked.

SECTION VII Potions & Brewing

Brewing Setup

Station: Potion Table (4 Wood Planks + 2 Glass). Base: Water Bottle (right-click Fat Cactus or water with Empty Bottle). Empty Bottle = 3 Glass + 1 Coal in Furnace. Splash potions = combine 2 of same potion. Effect Arrows = 1 Potion + 3 Arrows at Potion Table.

POTION	INGREDIENT	DURATION	UPGRADE	SPLASH
Strength	Water + Plum	30s	II (Moonstone)	Yes
Defense	Water + Pear	30s	II (Moonstone)	Yes
Speed	Water + Cracked Coconut	30s	II (Moonstone)	Yes
Jump	Water + Watermelon Slice	30s	II (Iron Bar)	Yes
Invisibility	Water + Melon Slice	30s	No	Yes
Haste	Water + Bowl of Cranberries	30s	II (Iron Bar)	Yes
Regeneration	Water + Cherry + Melon Slice	20s	II (Iron Bar)	Yes
Instant Healing	Water + Apple + Watermelon Slice	Instant	II (Iron Bar)	Yes
Instant Damage	Water + Cactus + Pumpkin Pie	Instant	II (Iron Bar)	Yes
Weakness	Water + Pumpkin	30s	II (Iron Bar)	Yes
Poison	Water + Cactus	15s	II (Iron Bar)	Yes
Slowness	Water + Poppy + Bread	20s	II (Iron Bar)	Yes
Shield	Water + Corn	30s	II (Iron Bar)	Yes
Heat Resistance	Water + Chili Pepper	60s	II (Iron Bar)	Yes
Aura XP	Water + Aura XP Orb	30s	II (Iron Bar)	Yes
Mining Yield	Water + Beetroot	30s	II (Iron Bar)	Yes
Brain Rot	Water + Rotten Brain	15s	No	Yes
Knockback	Water + Mango	20s	No	Yes
X-Ray Vision	Water + Carrot	30s	No	Yes
Double Jump	Water + Banana	30s	No	Yes
Milk	Direct from Cows	Instant	No	No

1. Ores — Depth & Drop Table

ORE	Y-LEVEL RANGE	BEST Y	MIN. PICKAXE	DROPS (NORMAL)	DROPS (MOONSTONE PICKAXE)
Coal	60 to -70	0 to 30	Wood	Coal	Coal Ore Block
Iron	0 to -80	-30 to -50	Stone	Iron Ore → Smelt	Iron Bar
Gold	-55 to -99	-70 to -85	Iron	Gold Ore → Smelt	Gold Bar
Diamond	-85 to -99	-90 to -99	Gold	Diamond	Diamond
Moonstone	-85 and below	-95 to -99	Diamond	Moonstone	Moonstone

2. Wood Types (12 Trees)

Each tree type produces: Log, Planks (4 per log), Leaves, Saplings. Seven types also have Fruity Leaves.

TREE	BIOME	FRUITY DROP	WOOD COLOR	SAPLING REGROW
Maple	All grass biomes	Apple	Light brown	~60s
Aspen	Aspen Forest	—	White/tan	~60s
Plum	Plum Forest	Plum	Dark brown	~60s
Pine	Pine/Cedar Forest	Pine Cone	Light tan	~90s
Cedar	Cedar Forest	—	Reddish	~90s
Jungle	Jungle	—	Dark red	~90s
Palm	Desert edges	Coconut	Tan	~120s
Mango	Tropical	Mango	Green-brown	~90s
Pear	Pear Forest	Pear	Light	~60s
Cherry	Cherry Forest	Cherry	Pink-brown	~60s
Spectral	Frozen Badlands	—	Blue-white	~120s

3. Crafted & Utility Blocks

- **Building:** Baked Clay (16 colors), Concrete, Bricks, Glass, Sandstone, Chalk, Dark Red Stone, Ice Bricks, Wool (16 colors), Banners (16 colors), Ceramics, Golden Decoration.
- **Utility:** Workbench, Artisan Bench, Furnace, Potion Table, Enchantment Table, Chest, Iron Chest, Bed, Doors & Trapdoors, Board, Code Block, Protector, Mob Spawner, Auto-Miner.
- **Slabs:** Grass, Dirt, Chalk, Chalk Brick, Concrete.
- **Farming:** Tilled Soil + Seeds for Wheat, Beetroot, Corn, Banana, Chili Pepper, Cotton, Carrot, Potato, Rice.

4. Processed Materials

- **Bars & Gems:** Coal, Iron Bar, Gold Bar, Diamond, Moonstone, Iron Fragment, Gold Fragment, Golem Eye.
- **Blocks of:** Block of Iron, Block of Gold.
- **Crafting:** Wood Planks (any type), Sticks, Cotton, Wheat, Rice, Bone, Fur, Crystal, Glowing Mushroom.
- **Misc:** Compass, Boat, Ball, Book, Bucket, Empty Bottle, Bowl, Fertiliser, Hang Gliders, Saplings (10 types).

SECTION IX Enchanting

Applied at the Enchantment Table. Tiers 1 through 5. Higher tiers require more XP.

ENCHANTMENT	APPLIES TO	EFFECT
Damage I / II / III	Swords, Bows, Crossbows	+7/+10/+13 base damage (swords), +8/+12/+16 (bows)
Critical Damage	Swords	Increases crit multiplier from 1.35x to ~1.5x
Vertical Knockback	Swords	Launches targets upward on hit
Break Speed	All tools	Faster block breaking
Momentum	All tools	Consecutive breaks get progressively faster
Mining Yield	Pickaxes	Chance to double ore drops
Mining Aura	Pickaxes	Mines adjacent blocks in a small radius
Digging Aura	Spades	Digs adjacent soil/sand in a small radius
Lumber Aura	Axes	Chops adjacent wood blocks
Farming Yield	Hoes	Chance to double crop drops
Farming Aura	Hoes	Harvests adjacent crops

SECTION X Player Actions API

Runtime: QuickJS (synchronous JavaScript). No async/await, no browser APIs. Code runs when a player clicks a Code Block or adjacent Board. Variables reset between uses unless global. World Code runs server-side via F8 (lobby owners only).

api.setVelocity(id, x, y, z)

Sets absolute velocity. Overrides current movement.

Launch pad – fling player upward

```
api.setVelocity(playerId, 0, 25, 0);
api.sendMessage(playerId, "Launched!", {color: "yellow"});
```

Speed boost forward

```
api.setVelocity(playerId, 15, 2, 0);
api.sendMessage(playerId, "Speed boost!", {color: "cyan"});
```

Freeze player in place

```
api.setVelocity(playerId, 0, 0, 0);
```

api.applyImpulse(id, x, y, z)

Adds to current velocity (doesn't override). Good for knockback.

Knockback on click

```
api.applyImpulse(playerId, -8, 5, 0);
api.sendMessage(playerId, "Pushed back!", {color: "red"});
```

Gentle upward bounce

```
api.applyImpulse(playerId, 0, 10, 0);
```

Random direction scatter

```
var rx = (Math.random() - 0.5) * 20;
var rz = (Math.random() - 0.5) * 20;
api.applyImpulse(playerId, rx, 8, rz);
```

api.setCameraDirection(id, [x, y, z])

Forces the player's camera to look in a direction.

Force look straight down

```
api.setCameraDirection(playerId, [0, -1, 0]);
```

Force look at sky

```
api.setCameraDirection(playerId, [0, 1, 0]);
```

api.forceRespawn(id)

Teleports player to spawn. Doesn't kill them.

Respawn with a warning

```
api.sendMessage(playerId, "Returning to spawn...", {color: "yellow"});  
api.forceRespawn(playerId);
```

Respawn all players

```
var ids = api.getPlayerIds(50);  
for (var i = 0; i < ids.length; i++) {  
  api.forceRespawn(ids[i]);  
}
```

api.setPlayerPose(id, pose)

Poses: "zombie", "driving".

Zombie walk on enter

```
api.setPlayerPose(playerId, "zombie");  
api.sendMessage(playerId, "You feel strange...", {color: "green"});
```

Driving pose for vehicle zone

```
api.setPlayerPose(playerId, "driving");  
api.applyEffect(playerId, "Speed", 999, {inbuiltLevel: 2});
```

api.setPlayerOpacity(id, value)

Value from 0 (invisible) to 1 (fully opaque).

Ghost mode – semi-transparent

```
api.setPlayerOpacity(playerId, 0.3);  
api.sendMessage(playerId, "You are now a ghost!", {color: "gray"});
```

Restore full visibility

```
api.setPlayerOpacity(playerId, 1);
```

api.kickPlayer(id, reason)

Kick for entering restricted area

```
api.kickPlayer(playerId, "You entered a restricted zone!");
```

api.setClientOption(id, key, value)

Client-side settings. Key options: "airJumpCount", "music", "movementSpeed".

Triple jump

```
api.setClientOption(playerId, "airJumpCount", 3);
```

Play background music

```
api.setClientOption(playerId, "music", "Butterfly Effect");
```

Disable air jumps

```
api.setClientOption(playerId, "airJumpCount", 0);
```

api.changePlayerIntoSkin(id, part, skin)

Skins: "farmer", "Wizard", "Zombie", "portal_mage", "trader".

Transform into wizard on click

```
api.changePlayerIntoSkin(playerId, "head", "Wizard");  
api.sendMessage(playerId, "You are now a Wizard!", {color: "purple"});
```

Random NPC skin

```
var skins = ["farmer", "Wizard", "Zombie", "trader"];  
var pick = skins[Math.floor(Math.random() * skins.length)];  
api.changePlayerIntoSkin(playerId, "head", pick);
```

api.scalePlayerMeshNodes(id, nodeScales)

Giant head mode

```
api.scalePlayerMeshNodes(playerId, {  
  HeadMesh: [3, 3, 3]  
});
```

Tiny body, big arms

```
api.scalePlayerMeshNodes(playerId, {
  TorsoNode: [0.5, 0.5, 0.5],
  ArmRightMesh: [2, 2, 2],
  ArmLeftMesh: [2, 2, 2]
});
```

SECTION XI Messaging API

api.sendMessage(id, text, options)

Send a chat message to one player. Colors: "red", "green", "blue", "cyan", "yellow", "purple", "gray", "white".

Welcome message

```
api.sendMessage(playerId, "Welcome to the arena!", {color: "cyan"});
```

Warning message

```
api.sendMessage(playerId, "You are entering the danger zone!", {color: "red"});
```

Show player their coordinates

```
var pos = api.getPosition(playerId);
api.sendMessage(playerId, "Position: " + Math.floor(pos[0]) + ", " + Math.floor(pos[1])
```

api.broadcastMessage(text, options)

Send a message to all players in the lobby.

Server announcement

```
api.broadcastMessage("The event starts in 10 seconds!", {color: "yellow"});
```

Announce a kill

```
var killerName = api.getEntityName(killerId);
var victimName = api.getEntityName(victimId);
api.broadcastMessage(killerName + " eliminated " + victimName + "!", {color: "red"});
```

api.sendFlyingMiddleMessage(id, lines, duration)

Big centered text on screen. Lines is an array. Duration in ticks (60 = 1s).

Countdown display

```
api.sendFlyingMiddleMessage(playerId, ["3"], 60);
```

Multi-line title screen

```
api.sendFlyingMiddleMessage(playerId, ["BOSS FIGHT", "Defeat the Cave Golem"], 180);
```

Victory announcement to all

```
var ids = api.getPlayerIds(50);
for (var i = 0; i < ids.length; i++) {
  api.sendFlyingMiddleMessage(ids[i], ["GAME OVER", "Team Red Wins!"], 300);
}
```

api.sendTopRightHelper(id, icon, text, options)

Shows a helper tooltip in the top-right corner.

Quest objective display

```
api.sendTopRightHelper(playerId, "Diamond Sword", "Collect 5 diamonds", {
  color: "cyan", duration: 300, height: 60, width: 200, fontSize: 14
});
```

Zone entry notification

```
api.sendTopRightHelper(playerId, "Compass", "You entered the PvP zone", {
  color: "red", duration: 120
});
```

SECTION XII Entity & Block API

api.getPosition(id) / api.setPosition(id, x, y, z)

Teleport player to specific coordinates

```
api.setPosition(playerId, 100, 65, -200);
api.sendMessage(playerId, "Teleported!", {color: "cyan"});
```

Teleport to another player

```
var ids = api.getPlayerIds(50);
if (ids.length > 1) {
  var target = ids[0] === playerId ? ids[1] : ids[0];
  var pos = api.getPosition(target);
  api.setPosition(playerId, pos[0], pos[1] + 2, pos[2]);
}
```

Move player up 10 blocks

```
var pos = api.getPosition(playerId);
api.setPosition(playerId, pos[0], pos[1] + 10, pos[2]);
```

api.getPlayerIds(max) / api.getNumPlayers(max)

Count online players

```
var count = api.getNumPlayers(100);
api.sendMessage(playerId, "Players online: " + count, {color: "green"});
```

List all player names

```
var ids = api.getPlayerIds(50);
var names = [];
for (var i = 0; i < ids.length; i++) {
  names.push(api.getEntityName(ids[i]));
}
api.sendMessage(playerId, "Online: " + names.join(", "), {color: "white"});
```

api.setBlock(x, y, z, type) / api.getBlock(x, y, z)

Place a gold block at player position

```
var pos = api.getPosition(playerId);
api.setBlock(Math.floor(pos[0]), Math.floor(pos[1]) - 1, Math.floor(pos[2]), "Block of
```

Check what block is below the player

```
var pos = api.getPosition(playerId);
var block = api.getBlock(Math.floor(pos[0]), Math.floor(pos[1]) - 1, Math.floor(pos[2])
api.sendMessage(playerId, "Standing on: " + block, {color: "gray"});
```

Build a 5-block tall pillar

```
for (var y = 0; y < 5; y++) {  
    api.setBlock(thisPos[0], thisPos[1] + y, thisPos[2] + 2, "Stone");  
}
```

api.setBlockRect(pos1, pos2, type)

Fills a rectangular area with a block type.

Create a 10x1x10 stone platform

```
api.setBlockRect(  
    [thisPos[0] - 5, thisPos[1], thisPos[2] - 5],  
    [thisPos[0] + 5, thisPos[1], thisPos[2] + 5],  
    "Stone"  
);
```

Clear a room (fill with air)

```
api.setBlockRect(  
    [10, 60, 10],  
    [20, 65, 20],  
    "Air"  
);
```

Build glass walls around an arena

```
// North wall  
api.setBlockRect([0, 60, 0], [20, 65, 0], "Glass");  
// South wall  
api.setBlockRect([0, 60, 20], [20, 65, 20], "Glass");  
// East wall  
api.setBlockRect([20, 60, 0], [20, 65, 20], "Glass");  
// West wall  
api.setBlockRect([0, 60, 0], [0, 65, 20], "Glass");
```

api.getBlockCoordinatesPlayerStandingOn(id) /

api.getBlockTypesPlayerStandingOn(id)

Lava floor trap

```
var blocks = api.getBlockTypesPlayerStandingOn(playerId);  
if (blocks && blocks.indexOf("Red Wool") !== -1) {  
    api.applyHealthChange(playerId, -15);  
    api.sendMessage(playerId, "The floor burns!", {color: "red"});  
}
```

SECTION XIII Health, Damage & Effects API

api.getHealth(id) / api.setHealth(id, val) / api.applyHealthChange(id, amt)

Healing shrine

```
api.setHealth(playerId, 100);
api.sendMessage(playerId, "Fully healed!", {color: "green"});
api.playParticleEffect({type: "simple", pos: api.getPosition(playerId), texture: "heart"}
```

Poison trap – deal 20 damage

```
api.applyHealthChange(playerId, -20);
api.sendMessage(playerId, "Poisoned! -20 HP", {color: "red"});
```

Show current health

```
var hp = api.getHealth(playerId);
api.sendMessage(playerId, "Your HP: " + hp + "/100", {color: hp > 50 ? "green" : "red"});
```

api.killLifeform(id) / api.isAlive(id)

Instant kill trap

```
api.killLifeform(playerId);
api.broadcastMessage(api.getEntityName(playerId) + " fell into the void!", {color: "red"});
```

Check if player is alive before rewarding

```
if (api.isAlive(playerId)) {
    api.giveItem(playerId, "Diamond", 5);
    api.sendMessage(playerId, "Reward: 5 Diamonds!", {color: "cyan"});
}
```

api.getCurrentKillstreak(id) / api.clearKillstreak(id)

Killstreak rewards

```
var streak = api.getCurrentKillstreak(playerId);
if (streak >= 5) {
    api.giveItem(playerId, "Diamond Sword", 1);
    api.broadcastMessage(api.getEntityName(playerId) + " is on a 5-kill streak!", {color: "green"});
}
if (streak >= 10) {
    api.applyEffect(playerId, "Strength", 600, {inbuiltLevel: 2});
    api.broadcastMessage("UNSTOPPABLE!", {color: "red"});
}
```


api.applyEffect(id, effectName, duration, options)

Duration in ticks (60 ticks = 1 second). Valid effects: Damage, Frozen, Speed, Jump, Invisibility, Haste, Regeneration, Weakness, Poison, Shield, Heat Resistance, Brain Rot, Double Jump, Knockback, X-Ray Vision, Mining Yield, Strength, Defense.

Speed boost for 10 seconds

```
api.applyEffect(playerId, "Speed", 600, {inbuiltLevel: 2});
api.sendMessage(playerId, "Speed II for 10s!", {color: "cyan"});
```

Freeze a player for 5 seconds

```
api.applyEffect(playerId, "Frozen", 300, {inbuiltLevel: 1});
api.sendMessage(playerId, "You are frozen!", {color: "blue"});
```

Full combat buff stack

```
api.applyEffect(playerId, "Strength", 600, {inbuiltLevel: 1});
api.applyEffect(playerId, "Speed", 600, {inbuiltLevel: 1});
api.applyEffect(playerId, "Regeneration", 600, {inbuiltLevel: 1});
api.applyEffect(playerId, "Defense", 600, {inbuiltLevel: 1});
api.sendMessage(playerId, "Full buff activated!", {color: "purple"});
```

Invisibility cloak (30 seconds)

```
api.applyEffect(playerId, "Invisibility", 1800, {inbuiltLevel: 1});
api.setPlayerOpacity(playerId, 0.1);
api.sendMessage(playerId, "You vanished!", {color: "gray"});
```

SECTION XIV Inventory, Crafting & Shops API

api.giveItem(id, itemName, amount, options?)

Optional: `{customDisplayName: "Excalibur", customAttributes: {damage: 99}}`

Starter kit

```
api.giveItem(playerId, "Iron Sword", 1);
api.giveItem(playerId, "Iron Pickaxe", 1);
api.giveItem(playerId, "Bread", 32);
api.giveItem(playerId, "Wood Planks", 64);
api.sendMessage(playerId, "Starter kit received!", {color: "green"});
```

Custom named weapon

```
api.giveItem(playerId, "Diamond Sword", 1, {
  customDisplayName: "Blade of the Ancients"
});
```

Random loot drop

```
var loot = ["Diamond", "Gold Bar", "Iron Bar", "Coal", "Apple"];
var item = loot[Math.floor(Math.random() * loot.length)];
var amt = Math.floor(Math.random() * 5) + 1;
api.giveItem(playerId, item, amt);
api.sendMessage(playerId, "Loot: " + amt + "x " + item, {color: "yellow"});
```

api.removeItemName(id, item, amount)

Pay 10 gold to enter

```
api.removeItemName(playerId, "Gold Bar", 10);
api.sendMessage(playerId, "Paid 10 Gold. Welcome!", {color: "yellow"});
```

api.getHeldItem(id) / api.setItemSlot(id, slot, item)

Check held weapon and react

```
var held = api.getHeldItem(playerId);
if (held === "Diamond Sword") {
  api.sendMessage(playerId, "A worthy blade.", {color: "cyan"});
} else {
  api.sendMessage(playerId, "You need a Diamond Sword!", {color: "red"});
}
```

Force equip item in hotbar slot 0

```
api.setItemSlot(playerId, 0, "Knight Sword");
```

api.clearInventory(id)

Reset inventory for new round

```
api.clearInventory(playerId);
api.sendMessage(playerId, "Inventory cleared for round 2.", {color: "yellow"});
```

api.createShopItem(category, key, options)

Create a weapon shop

```
api.createShopItem("Weapons", "iron_sword", {
  name: "Iron Sword",
  icon: "Iron Sword",
  price: 50,
  currency: "Gold Bar"
});
api.createShopItem("Weapons", "diamond_sword", {
  name: "Diamond Sword",
  icon: "Diamond Sword",
  price: 200,
  currency: "Gold Bar"
});
```

api.getStandardChestItems(pos) / api.setStandardChestItemSlot(pos, slot, item)

Stock a loot chest

```
api.setStandardChestItemSlot([10, 62, 10], 0, "Diamond");
api.setStandardChestItemSlot([10, 62, 10], 1, "Gold Bar");
api.setStandardChestItemSlot([10, 62, 10], 2, "Bread");
```

api.editItemCraftingRecipes(id, item, recipe)

Make diamond sword cheaper

```
api.editItemCraftingRecipes(playerId, "Diamond Sword", {
  ingredients: [{item: "Diamond", amount: 1}, {item: "Stick", amount: 1}]
});
```

SECTION XV Visuals, Particles & Mesh API

api.playParticleEffect(options)

Textures: `bubble`, `critical_hit`, `drift`, `effect_5`, `generic_2`, `glint`, `heart`, `soul_0`, `square_particle`, `smoke`.

Heart particles at player

```
var pos = api.getPosition(playerId);
api.playParticleEffect({
  type: "simple",
  pos: pos,
  texture: "heart"
});
```

Smoke explosion

```
api.playParticleEffect({
  type: "simple",
  pos: [thisPos[0], thisPos[1] + 1, thisPos[2]],
  texture: "smoke"
});
```

Critical hit sparks

```
var pos = api.getPosition(playerId);
api.playParticleEffect({type: "simple", pos: pos, texture: "critical_hit"});
api.playParticleEffect({type: "simple", pos: pos, texture: "glint"});
```

api.updateEntityNodeMeshAttachment(id, node, type, opts, pos, rot)

Nodes: `TorsoNode` , `HeadMesh` , `ArmRightMesh` , `ArmLeftMesh` , `LegLeftMesh` , `LegRightMesh` .

Attach a gold crown to head

```
api.updateEntityNodeMeshAttachment(
  playerId, "HeadMesh", "BloxdBlock",
  {blockType: "Block of Gold", scale: 0.3},
  [0, 0.5, 0], [0, 0, 0]
);
```

Shield on left arm

```
api.updateEntityNodeMeshAttachment(
  playerId, "ArmLeftMesh", "BloxdBlock",
  {blockType: "Iron Block", scale: 0.5},
  [0.3, 0, 0], [0, 0, 0]
);
```

api.progressBarUpdate(id, progress, duration)

Loading bar at 50%

```
api.progressBarUpdate(playerId, 0.5, 120);
```

Complete the progress bar

```
api.progressBarUpdate(playerId, 1.0, 60);  
api.sendMessage(playerId, "Task complete!", {color: "green"});
```

SECTION XVI Mob Spawning API

api.attemptSpawnMob(type, x, y, z, options)

Mob types: "Cave Golem", "Sheep", "Pig", "Cow", "Zombie", "Skeleton", "Stag", "Chicken", "Gorilla", "Draugr Knight".

Spawn a zombie near the player

```
var pos = api.getPosition(playerId);  
api.attemptSpawnMob("Zombie", pos[0] + 5, pos[1], pos[2], {  
    playSoundOnSpawn: true  
});
```

Spawn a named boss golem

```
api.attemptSpawnMob("Cave Golem", thisPos[0], thisPos[1] + 2, thisPos[2], {  
    name: "Iron Guardian",  
    variation: "iron",  
    playSoundOnSpawn: true  
});  
api.broadcastMessage("The Iron Guardian has awakened!", {color: "red"});
```

Spawn a rainbow sheep flock

```
var colors = ["red", "blue", "green", "yellow", "pink", "purple"];  
for (var i = 0; i < colors.length; i++) {  
    api.attemptSpawnMob("Sheep", thisPos[0] + i * 2, thisPos[1] + 1, thisPos[2], {  
        variation: colors[i]  
    });  
}
```

Zombie horde wave spawner

```
var waveSize = 8;
for (var i = 0; i < waveSize; i++) {
    var angle = (i / waveSize) * Math.PI * 2;
    var x = thisPos[0] + Math.cos(angle) * 10;
    var z = thisPos[2] + Math.sin(angle) * 10;
    api.attemptSpawnMob("Zombie", x, thisPos[1] + 1, z, {playSoundOnSpawn: true});
}
api.broadcastMessage("Wave incoming! " + waveSize + " zombies!", {color: "red"});
```

Spawn a farm – mixed passive mobs

```
var animals = ["Cow", "Pig", "Sheep", "Chicken"];
for (var i = 0; i < animals.length; i++) {
    api.attemptSpawnMob(animals[i],
        thisPos[0] + Math.floor(Math.random() * 6),
        thisPos[1] + 1,
        thisPos[2] + Math.floor(Math.random() * 6), {});
}
```

SECTION XVII Utilities API

api.log(msg) / console.log(msg)

Debug player position

```
var pos = api.getPosition(playerId);
api.log("Player pos: " + pos[0] + "," + pos[1] + "," + pos[2]);
```

Log player count

```
console.log("Total players: " + api.getNumPlayers(100));
```

api.now()

Cooldown timer

```
var lastUse = 0;
var cooldown = 5000; // 5 seconds
if (api.now() - lastUse < cooldown) {
    api.sendMessage(playerId, "On cooldown!", {color: "red"});
} else {
    lastUse = api.now();
    api.sendMessage(playerId, "Ability used!", {color: "green"});
}
```

api.checkValid(id)

Safe entity operation

```
if (api.checkValid(playerId)) {
  api.giveItem(playerId, "Diamond", 1);
} else {
  api.log("Player no longer valid!");
}
```

SECTION XVIII World Code Callbacks (64 Total)

World Code runs globally via F8 (lobby owners only). Callbacks fire on events. Each callback function receives an `event` object with relevant data.

Core Lifecycle

tick

Fires every game tick (~16ms). Use for continuous game logic.

Shrinking border every 30 seconds

```
var borderSize = 100;
var tickCount = 0;
world.onTick = function() {
  tickCount++;
  if (tickCount % 1800 === 0 && borderSize > 20) {
    borderSize -= 5;
    api.broadcastMessage("Border shrinking! Size: " + borderSize, {color: "red"});
  }
  var ids = api.getPlayerIds(50);
  for (var i = 0; i < ids.length; i++) {
    var pos = api.getPosition(ids[i]);
    if (Math.abs(pos[0]) > borderSize || Math.abs(pos[2]) > borderSize) {
      api.applyHealthChange(ids[i], -5);
    }
  }
};
```

Player Join / Leave

onPlayerJoin / onPlayerLeave

Welcome kit + join announcement

```
world.onPlayerJoin = function(e) {
  var name = api.getEntityName(e.entity);
  api.broadcastMessage(name + " joined the game!", {color: "green"});
  api.giveItem(e.entity, "Stone Sword", 1);
  api.giveItem(e.entity, "Bread", 16);
  api.sendFlyingMiddleMessage(e.entity, ["Welcome!", "Survive the night."], 180);
};
```

Leave announcement

```
world.onPlayerLeave = function(e) {
  var name = api.getEntityName(e.entity);
  api.broadcastMessage(name + " left.", {color: "gray"});
};
```

Chat & Commands

playerCommand / onPlayerChat

Custom chat commands

```
world.playerCommand = function(e) {
  var cmd = e.command;
  if (cmd === "heal") {
    api.setHealth(e.entity, 100);
    api.sendMessage(e.entity, "Healed!", {color: "green"});
  } else if (cmd === "kit") {
    api.giveItem(e.entity, "Diamond Sword", 1);
    api.giveItem(e.entity, "Steak", 64);
  } else if (cmd === "spawn") {
    api.forceRespawn(e.entity);
  } else if (cmd === "players") {
    api.sendMessage(e.entity, "Online: " + api.getNumPlayers(100), {color: "cyan"});
  }
};
```

Chat filter

```
world.onPlayerChat = function(e) {
  api.log("Chat: " + api.getEntityName(e.entity) + ": " + e.message);
};
```


Combat Callbacks

onPlayerAttack / onPlayerKilledOtherPlayer / onMobKilledPlayer

PvP kill tracking with rewards

```
var kills = {};  
world.onPlayerKilledOtherPlayer = function(e) {  
    var killer = e.entity;  
    var victim = e.targetEntity;  
    var killerName = api.getEntityName(killer);  
    var victimName = api.getEntityName(victim);  
  
    if (!kills[killer]) kills[killer] = 0;  
    kills[killer]++;  
  
    api.broadcastMessage(killerName + " killed " + victimName + " [" + kills[killer] + "  
    api.giveItem(killer, "Gold Bar", 5);  
  
    if (kills[killer] % 5 === 0) {  
        api.giveItem(killer, "Diamond", 3);  
        api.broadcastMessage(killerName + " reached " + kills[killer] + " kills!", {color:  
    }  
};
```

onPlayerDamagingOtherPlayer

Damage multiplier zone

```
world.onPlayerDamagingOtherPlayer = function(e) {  
    var pos = api.getPosition(e.entity);  
    if (pos[1] > 80) {  
        e.damage = e.damage * 2;  
        api.sendMessage(e.entity, "HIGH GROUND BONUS! 2x damage!", {color: "yellow"});  
    }  
};
```

onPlayerKilledMob / onMobDamagingPlayer

Mob kill rewards

```
world.onPlayerKilledMob = function(e) {  
    api.giveItem(e.entity, "Gold Bar", 2);  
    api.sendMessage(e.entity, "Mob killed! +2 Gold", {color: "yellow"});  
};
```

Block Interaction

onPlayerChangeBlock / onBlockStand

Anti-grief: prevent block breaking in spawn area

```
world.onPlayerChangeBlock = function(e) {  
  if (Math.abs(e.x) < 20 && Math.abs(e.z) < 20 && e.y < 70) {  
    e.cancel = true;  
    api.sendMessage(e.entity, "No building in spawn!", {color: "red"});  
  }  
};
```

Pressure plate: boost when standing on gold

```
world.onBlockStand = function(e) {  
  if (e.blockType === "Block of Gold") {  
    api.applyImpulse(e.entity, 0, 20, 0);  
  }  
};
```

Inventory Callbacks

onPlayerPickedUpItem / onPlayerDropItem

Log item pickups

```
world.onPlayerPickedUpItem = function(e) {  
  api.log(api.getEntityName(e.entity) + " picked up " + e.itemName);  
};
```

Prevent dropping diamonds

```
world.onPlayerDropItem = function(e) {  
  if (e.itemName === "Diamond") {  
    e.cancel = true;  
    api.sendMessage(e.entity, "Can't drop Diamonds!", {color: "red"});  
  }  
};
```

Crafting & Chests

onPlayerAttemptCraft / onPlayerCraft

Block crafting of certain items

```
world.onPlayerAttemptCraft = function(e) {  
  if (e.itemName === "Diamond Sword") {  
    e.cancel = true;  
    api.sendMessage(e.entity, "Diamond Swords must be earned!", {color: "red"});  
  }  
};
```

Crafting XP reward

```
world.onPlayerCraft = function(e) {  
  api.sendMessage(e.entity, "Crafted: " + e.itemName + " +10 XP", {color: "green"});  
};
```

Effects & Items

onPlayerPotionEffect / onPlayerUsedThrowable

Log potion usage

```
world.onPlayerPotionEffect = function(e) {  
  api.log(api.getEntityName(e.entity) + " used potion: " + e.effectName);  
};
```

Throwable impact notification

```
world.onPlayerUsedThrowable = function(e) {  
  api.broadcastMessage(api.getEntityName(e.entity) + " threw a projectile!", {color: "red"});  
};
```

Mob Lifecycle

onWorldSpawnMob / onMobDespawned

Track mob population

```
var mobCount = 0;
world.onWorldSpawnMob = function(e) {
  mobCount++;
  api.log("Mob spawned. Total: " + mobCount);
};
world.onMobDespawned = function(e) {
  mobCount--;
  api.log("Mob despawned. Total: " + mobCount);
};
```

Complete Callback Reference

CATEGORY	CALLBACKS
Core	<code>tick</code> , <code>onClose</code> , <code>doPeriodicSave</code>
Player Join/Leave	<code>onPlayerJoin</code> , <code>onPlayerLeave</code> , <code>onPlayerJump</code> , <code>onRespawnRequest</code>
Chat & Commands	<code>playerCommand</code> , <code>onPlayerChat</code>
Block Interaction	<code>onPlayerChangeBlock</code> , <code>onBlockStand</code> , <code>onWorldChangeBlock</code> , <code>onChunkLoaded</code> , <code>onPlayerRequestChunk</code>
Inventory	<code>onPlayerDropItem</code> , <code>onPlayerPickedUpItem</code> , <code>onPlayerSelectInventorySlot</code> , <code>onPlayerMoveItemOutOfInventory</code> , <code>onPlayerMoveInvenItem</code> , <code>onPlayerMoveItemIntoIdxs</code> , <code>onPlayerSwapInvenSlots</code> , <code>onPlayerMoveInvenItemWithAmt</code> , <code>onInventoryUpdated</code> , <code>onItemDropCreated</code>
Crafting & Chests	<code>onPlayerAttemptCraft</code> , <code>onPlayerCraft</code> , <code>onPlayerAttemptOpenChest</code> , <code>onPlayerOpenedChest</code> , <code>onChestUpdated</code>
Actions & Clicks	<code>onPlayerAttemptAltAction</code> , <code>onPlayerAltAction</code> , <code>onPlayerClick</code> , <code>onPlayerClickUp</code> , <code>onTouchscreenActionButton</code>
Combat	<code>onPlayerAttack</code> , <code>onPlayerDamagingOtherPlayer</code> , <code>onPlayerDamagingMob</code> , <code>onMobDamagingPlayer</code> , <code>onMobDamagingOtherMob</code> , <code>onAttemptKillPlayer</code> , <code>onPlayerKilledOtherPlayer</code> , <code>onMobKilledPlayer</code> , <code>onPlayerKilledMob</code> , <code>onMobKilledOtherMob</code>
Mob Lifecycle	<code>onPlayerAttemptSpawnMob</code> , <code>onWorldAttemptSpawnMob</code> , <code>onPlayerSpawnMob</code> , <code>onWorldSpawnMob</code> , <code>onWorldAttemptDespawnMob</code> , <code>onMobDespawned</code>
Effects & Items	<code>onPlayerPotionEffect</code> , <code>onPlayerUsedThrowable</code> , <code>onPlayerThrowableHitTerrain</code> , <code>onPlayerStartChargingItem</code> , <code>onPlayerFinishChargingItem</code> , <code>onPlayerBoughtShopItem</code>
Mesh & Entities	<code>onCreateBloxdMeshEntity</code> , <code>onEntityCollision</code> , <code>onPlayerDamagingMeshEntity</code> , <code>onPlayerBreakMeshEntity</code>
Settings & Tasks	<code>onClientOptionUpdated</code> , <code>onMobSettingUpdated</code> , <code>onTaskClaimed</code> , <code>onPlayerFinishQTE</code>

SECTION XIX Biomes — Detailed Guide

Grass Biomes

Plains Grass

Temperature: Temperate • Terrain: Flat to gently rolling

Surface: Grass Block, Dirt. Flat terrain with occasional gentle hills.

Trees: Maple (common). Sparse tree density.

Flora: Flowers (Poppy, Dandelion), Tall Grass, occasional Pumpkins.

Mobs: Cow, Pig, Sheep (any color), Chicken. High passive mob density.

Ores below: All standard ores at normal rates. Good for early base building due to flat terrain.

Notes: Best biome for new players. Open sightlines make hostile mob detection easy at night.

Forest Grass

Temperature: Temperate • Terrain: Hilly

Surface: Grass Block, Dirt. Moderate hills with dense tree cover.

Trees: Maple (dominant), mixed species at biome edges.

Flora: Flowers, Mushrooms (under trees), Watermelon, Melon. Tall Grass is common.

Mobs: Cow, Pig, Sheep, Stag, Zombies and Skeletons at night.

Special drops: Watermelon Slice (Jump Boost), Melon Slice (Invisibility) from wild plants.

Notes: Excellent wood supply. Dangerous at night due to obstructed views.

Aspen Forest Grass

Temperature: Cool Temperate • Terrain: Rolling hills

Surface: Grass Block, Dirt.

Trees: Aspen (exclusive). White/tan bark, distinctive look. Medium density.

Flora: Flowers, Tall Grass. No fruity leaves.

Mobs: Stag (more common here), Sheep, Cow.

Notes: Aspen wood planks have a unique pale tone, popular in building.

Plum Forest Grass

Temperature: Temperate • Terrain: Gentle hills

Surface: Grass Block, Dirt.

Trees: Plum (exclusive). Dark brown wood. Has Fruity Plum Leaves.

Flora: Flowers, Plum drops from Fruity Leaves.

Mobs: Passive mobs. Standard hostiles at night.

Special drops: Plum fruit → Damage I effect (8s). Key ingredient for Strength Potions.

Notes: A must-visit biome for combat players. Plums provide the only food-based damage boost.

Pine Forest Grass

Temperature: Cool • Terrain: Steep hills, tall trees

Surface: Grass Block, Dirt, Rocky Dirt patches.

Trees: Pine (tall conifers). Dense canopy. Fruity Pine Leaves drop Pine Cones.

Mobs: Stag, Sheep. Wolves in some variants.

Notes: Pine logs are tall, yielding more wood per tree. Difficult terrain for building.

Cedar Forest Grass

Temperature: Cool • Terrain: Mountainous

Surface: Grass Block, Dirt, Stone outcrops.

Trees: Cedar (reddish wood). Very tall trees. No fruity leaves.

Mobs: Standard passive. Skeletons more common at night in elevated areas.

Notes: Cedar wood has a distinctive red-brown color. Good for themed builds.

Cherry Forest Grass

Temperature: Temperate • Terrain: Gentle rolling

Surface: Grass Block, Dirt.

Trees: Cherry (pink-brown wood, pink leaves). Has Fruity Cherry Leaves.

Special drops: Cherry fruit → Regeneration (5s). Key ingredient for Regeneration Potions.

Mobs: Passive mobs at normal rates.

Notes: Highly sought after for the Regeneration effect. Beautiful biome for screenshots.

Autumn Maple Forest Grass

Temperature: Cool • Terrain: Hilly

Surface: Grass Block, Dirt.

Trees: Maple with orange/red/yellow autumn leaves. Fruity Maple Leaves drop Apples.

Mobs: Standard passive + Stag.

Notes: Identical resources to regular Forest but with seasonal-colored leaves.

Pear Forest Grass

Temperature: Temperate • Terrain: Flat to gently hilly

Surface: Grass Block, Dirt.

Trees: Pear (light wood). Has Fruity Pear Leaves.

Special drops: Pear fruit → Damage Reduction (13s). Key ingredient for Defense Potions.

Notes: Pears are the best defensive food in the game. Stock up here.

Jungle Grass

Temperature: Tropical • Terrain: Dense, multi-layered canopy

Surface: Grass Block, Dirt, Vines.

Trees: Jungle (dark red wood, very tall, 2x2 trunk). Dense undergrowth.

Flora: Tall Grass, Ferns, Mushrooms (under canopy), Cocoa plants.

Mobs: Gorilla (exclusive — drops Bananas), Pig, Chicken.

Special drops: Banana → Double Jump effect.

Notes: Only biome with Gorillas. Navigation is difficult. Bring an axe.

Meadow Grass

Temperature: Temperate • Terrain: Flat, open

Surface: Grass Block, Dirt.

Trees: Very sparse Maple.

Flora: Dense flowers, Tall Grass, Corn plants (natural).

Mobs: Sheep (high density), Cow, Chicken.

Special: Natural Corn plants give Corn → 20 HP shield.

Notes: Best biome for finding Corn and starting farms. Very open — safe at night.

Desert Biomes

Desert Desert

Temperature: Hot • Terrain: Flat with dunes

Surface: Sand, Red Sand, Sandstone (underground).

Trees: Palm at edges (Fruity Palm Leaves → Coconut). No trees in center.

Flora: Cactus, Dead Bushes. Fat Cactus (water source for bottles).

Mobs: No passive mobs. Zombies and Skeletons at night.

Special: Fat Cactus provides water for brewing. Cactus used in Poison/Instant Damage potions.

Notes: Dangerous for new players — no food, no wood. Palm trees at edges are vital.

Badlands Desert

Temperature: Hot • Terrain: Steep plateaus, mesas

Surface: Red Sand, Baked Clay (multiple colors naturally), Sandstone.

Trees: None.

Flora: Dead Bushes, Cactus.

Mobs: No passive mobs. Higher hostile mob density at night.

Ores: Gold ore appears at higher rates in Badlands (above y:0).

Notes: Best biome for gold farming. Natural Baked Clay is excellent for building.

Cactus Desert Desert

Temperature: Hot • Terrain: Flat

Surface: Sand.

Trees: None.

Flora: Dense Cactus clusters, Fat Cactus (common).

Notes: Best source of Cactus for Poison potions and Fat Cactus for water.

Canyon Desert

Temperature: Hot • Terrain: Deep ravines, cliff walls

Surface: Sand, Red Sand, Stone, Sandstone.

Trees: None at surface. Occasional shrubs on ledges.

Mobs: Skeletons (in shadows). No passive mobs.

Ores: Exposed ore veins visible in canyon walls — good for early mining without digging.

Notes: Fall damage risk. Look for exposed iron and coal in cliff faces.

Snow Biomes

Snowy Plains Snow

Temperature: Freezing • Terrain: Flat, snow-covered

Surface: Snow, Packed Snow, Ice. Dirt underneath snow layer.

Trees: Sparse Pine with snow-covered leaves.

Flora: None above snow.

Mobs: Stag (exclusive cold-weather spawn), Sheep (white/gray). Skeletons at night.

Special: Fur drops from Stag → Fur Armor (18% damage reduction chestplate).

Notes: Only biome for Fur Armor materials. Farming impossible on snow.

Frozen Badlands Snow

Temperature: Freezing • Terrain: Ice plateaus, frozen mesas

Surface: Packed Snow, Ice, Baked Clay under ice layer.

Trees: Spectral Trees (exclusive). Blue-white wood, ghostly appearance.

Flora: None. Glowing Mushrooms in ice caves.

Mobs: Draugr Knight (exclusive hostile — drops Knight Hearts), Stag, Skeleton.

Special: Knight Hearts + Diamond Sword = Knight Sword (43 base damage, strongest sword).

Ores: Moonstone more common here. Crystals found in ice formations.

Notes: The endgame biome. Come prepared with Diamond gear and food. Draugr Knights are tough.

Underground Biomes

Mountains Underground

Terrain: High elevation peaks, exposed stone

Surface: Stone, Grass at lower elevations, Snow at peaks.

Trees: Pine at treeline. None at peaks.

Mobs: Stag, Sheep. Cave Golem in mountain caves.

Ores: Exposed coal and iron in cliff faces. Normal distribution underground.

Notes: Natural caves are common. Watch for Cave Golems in deeper caverns.

Stony Caves / Andesite Caves / Granite Caves / Diorite Caves Underground

Depth: y:0 and below • Terrain: Underground caverns

Composition: Each cave type is dominated by its namesake stone (Andesite, Granite, or Diorite). Stony Caves use standard Stone.

Features: Stalactites, open caverns, underground water pools. Glowing Mushrooms provide dim light.

Mobs: Zombie, Skeleton (always spawned in darkness), Cave Golem (rare but deadly).

Ores: Higher ore density in cave walls. All ore types present at appropriate depths.

Notes: Bring torches. Cave Golems have 200 HP and deal 40 damage — don't fight without iron+ gear.

Rivers Underground

Terrain: Water channels cutting through biomes

Surface: Water, Sand, Clay, Dirt banks.

Flora: Sugar Cane, Lily Pads.

Mobs: Passive mobs on banks. No aquatic mobs.

Special: Clay is exclusive to river banks. Used in Baked Clay crafting.

Notes: Rivers bisect biomes and provide natural Clay. Use Boats for fast travel.

SECTION XX

Mobs — Detailed Stats

Passive Mobs

Cow

Passive

HEALTH	40 HP	DAMAGE	o (never attacks)
SPEED	Slow	SPAWN BIOMES	Plains, Forest, Meadow
DROPS	Raw Beef (1-2), Bone (0-1)	SPECIAL	Right-click with Empty Bottle → Milk (clears all effects)

Cows are the primary source of Milk, which clears all potion effects (positive and negative). They also provide the best raw meat in the game — Steak heals 28 HP when cooked. Found in most grass biomes but most common in Plains and Meadow.

Pig

Passive

HEALTH	30 HP	DAMAGE	o
SPEED	Medium	SPAWN BIOMES	Plains, Forest, Jungle
DROPS	Raw Porkchop (1-2)	SPECIAL	Slightly faster than Cows, harder to chase

Cooked Porkchop heals 25 HP in 1s, making Pigs an efficient food source. They tend to run away faster than Cows when hit.

Sheep Passive

HEALTH	25 HP	DAMAGE	0
SPEED	Slow	SPAWN BIOMES	All grass biomes, Snowy Plains
DROPS	Raw Mutton (1), Wool (1-2, matches color)	VARIANTS	16 colors: White (default), Black, Red, Orange, Pink, Purple, Yellow, Blue, Brown, Cyan, Gray, Green, Light Blue, Light Gray, Lime, Magenta
SPECIAL	Use Shears for Wool without killing. Wool regrows.		

White sheep are the most common (~80% of spawns). Colored sheep are rarer, with Pink and Magenta being the rarest natural spawns. Sheep can be sheared repeatedly, making them a renewable wool source. Each color of wool can be used for building. 16 wool colors × armor pieces = 80 colored wood armor variants.

Stag Passive

HEALTH	35 HP	DAMAGE	0
SPEED	Fast (runs quickly when hit)	SPAWN BIOMES	Aspen Forest, Pine Forest, Snowy Plains, Mountains
DROPS	Raw Venison (1-2), Fur (1), Bone (0-1)	SPECIAL	Only source of Fur (used for Fur Armor)

Stags are the only mob that drops Fur, required for Fur Armor (18% chestplate DR). They're fast runners, so use ranged weapons or corner them against terrain. Most common in cold biomes. Venison heals only 16 HP when cooked — not the best food, but Fur makes hunting them worthwhile.

Chicken Passive

HEALTH	10 HP	DAMAGE	0
SPEED	Slow	SPAWN BIOMES	Plains, Forest, Jungle, Meadow
DROPS	Feathers (1-2), Bone (0-1)	SPECIAL	Takes no fall damage (glides down). Very low HP.

Chickens are the weakest mob. Their feathers are used in Arrow crafting. They slowly glide when falling, taking no fall damage. One-hit kill with any weapon.

Hostile Mobs

Zombie Hostile

HEALTH	80 HP	DAMAGE	18 per hit
SPEED	Slow (slightly slower than player walk)	SPAWN	Night (all biomes), Caves (always), Dark areas
DROPS	Rotten Brain (1), Bone (0-2)	AGGRO RANGE	~16 blocks
SPECIAL	Burns in sunlight. Attacks in groups.		

The most common hostile mob. Zombies are slow but deal moderate damage and often spawn in groups of 2-4. They burn in direct sunlight at dawn. Rotten Brain is used to brew Brain Rot potions. A Stone Sword or better can handle them easily. They can break down doors in some modes.

Skeleton Hostile

HEALTH	60 HP	DAMAGE	12-24 per arrow (varies with range)
SPEED	Medium (strafes while shooting)	SPAWN	Night (all biomes), Caves, Dark areas
DROPS	Bone (1-2), Arrows (0-3)	AGGRO RANGE	~24 blocks
SPECIAL	Ranged attack (bow). Burns in sunlight. Retreats to maintain distance.		

More dangerous than Zombies due to ranged attacks. Skeletons strafe left and right while shooting arrows, making them hard to approach. They have less HP than Zombies (60 vs 80) but their range compensates. Close the gap quickly with sprinting. Their arrow drops save crafting resources. They also burn in sunlight.

Cave Golem Boss-Tier

HEALTH	200 HP (Iron variant: 300 HP)	DAMAGE	40 per hit (Iron: 55 per hit)
SPEED	Slow (charges when attacking)	SPAWN	Deep caves (y: -40 and below), Mountains
DROPS	Golem Eye (1-2), Moonstone (0-1), Iron Fragment (2-4)	AGGRO RANGE	~12 blocks
VARIANTS	Default (stone), Iron (darker, stronger, rarer)	SPECIAL	Ground pound AoE attack. Immune to knockback. Does not burn in sunlight.

The strongest non-boss mob. Cave Golems are found deep underground and hit extremely hard. The Iron variant has 300 HP and deals 55 damage — a two-hit kill on unarmored players. They have a ground-pound AoE attack that damages all nearby players. Golem Eyes are needed for the Golem Pickaxe. Fight with Diamond armor, bring food (Bowl of Rice for fast healing), and keep distance with bows. They cannot be knocked back, so don't try melee-only without armor.

Special Mobs

Draugr Knight

Hostile

Boss-Tier

HEALTH	250 HP	DAMAGE	35 per hit (sword attacks)
SPEED	Medium-Fast	SPAWN	Frozen Badlands (exclusive)
DROPS	Knight Heart (1-2), Bone (2-3), Iron Bar (1-2)	AGGRO RANGE	~20 blocks
SPECIAL	Sword attacks with combos. Can block. Spawns with armor.		

The endgame combat mob. Found only in Frozen Badlands. Draugr Knights attack with sword combos and can block incoming attacks, making timing important. Knight Hearts (1-2 dropped) are used to craft the Knight Sword (Diamond Sword + 3 Knight Hearts = 43 base damage, the strongest sword in the game). Bring Diamond armor, a Diamond Sword, and plenty of healing food. Bows are effective for chipping HP before melee engagement.

Gorilla

Special

HEALTH	100 HP	DAMAGE	25 per hit (becomes aggressive if attacked)
SPEED	Fast when aggressive	SPAWN	Jungle (exclusive)
DROPS	Banana (2-4)	AGGRO RANGE	~8 blocks (only when provoked)
SPECIAL	Neutral — only attacks if hit first. High knockback attacks.		

Gorillas are neutral mobs — they won't attack unless provoked. Once angered, they're dangerous with 25 damage and high knockback. They're the only source of Bananas, which grant the Double Jump effect. Found exclusively in Jungles. You can farm Bananas from Banana plants (grown from seeds) as an alternative to fighting Gorillas.

SECTION XXI Music Tracks

Playable via `api.setClientOption(id, "music", "trackName")`.

Adigold

- A Place To Be Free
- Butterfly Effect
- Dreamless Sleep
- Frozen Pulse
- Frozen Skies
- Healing Thoughts
- Here Forever
- Just a Little Hope
- Just Like Heaven
- Memories Remain
- Place To Be
- The Riverside
- The Wonder
- Vetrar (Cut B)

Juhani Junkala [Retro Game Music Pack]

- Ending
- Level 1
- Level 2
- Level 3
- Title Screen

Mojo Productions

- Pirates
- Sneaky Jazz
- The Sneaky
- The Sneaky Jazz

Others

- Awkward Comedy Quirky
- battle-ship-111902
- cdk-Silence-Await
- corsairs-studiokolomna
- ghost-Reverie-small-theme
- happy
- Heroic-Demise-New

- I-am-the-Sea-The-Room-4
- LonePeakMusic-Highway-1
- progress
- raise-the-sails-152124
- ramblinglibrarian-I-Have-Often-T
- TownTheme

Play a track by name

```
api.setClientOption(playerId, "music", "Frozen Pulse");
```

Random ambient music

```
var tracks = ["Dreamless Sleep", "Healing Thoughts", "The Riverside", "Frozen Skies"];  
var pick = tracks[Math.floor(Math.random() * tracks.length)];  
api.setClientOption(playerId, "music", pick);  
api.sendMessage(playerId, "Now playing: " + pick, {color: "cyan"});
```

Battle music for all players

```
var ids = api.getPlayerIds(50);  
for (var i = 0; i < ids.length; i++) {  
    api.setClientOption(ids[i], "music", "Heroic-Demise-New");  
}
```

